

Precipitation

The build-up season in the Top End

The late-build-up season in **Darwin, NT**, is a period of oppressive heat and humidity. The temperature only rises a few degrees, but the humidity skyrockets. It does not rain as often as the wet season, but afternoon thunderstorms are frequent:

- List **three atmospheric conditions** that must be satisfied before any of these storms can produce precipitation.
- For each condition, explain (≈ 2 sentences) why failure of that condition would prevent rainfall – even under high relative humidity.

Warm vs Mixed-Phase Clouds

Convective downpours over **Singapore** in March are dominated by warm-rain processes, whereas December snow-storms in the **Scottish Highlands** rely heavily on the Bergeron ice-crystal mechanism.

- Identify the **dominant droplet-growth mechanism** operating in each region.
- Describe how cloud-temperature structure and up-draught strength favour those mechanisms.
- State one consequence for **raindrop/snow-crystal size distribution at the ground** in each case.

Sierra Nevada Transect

Fig. 1 shows a west-to-east cross-section across California's San Jose from the Pacific coast to Bishop. On your copy of the figure, mark:

1. the zone of **maximum orographic precipitation**.
2. the **rain-shadow region**.

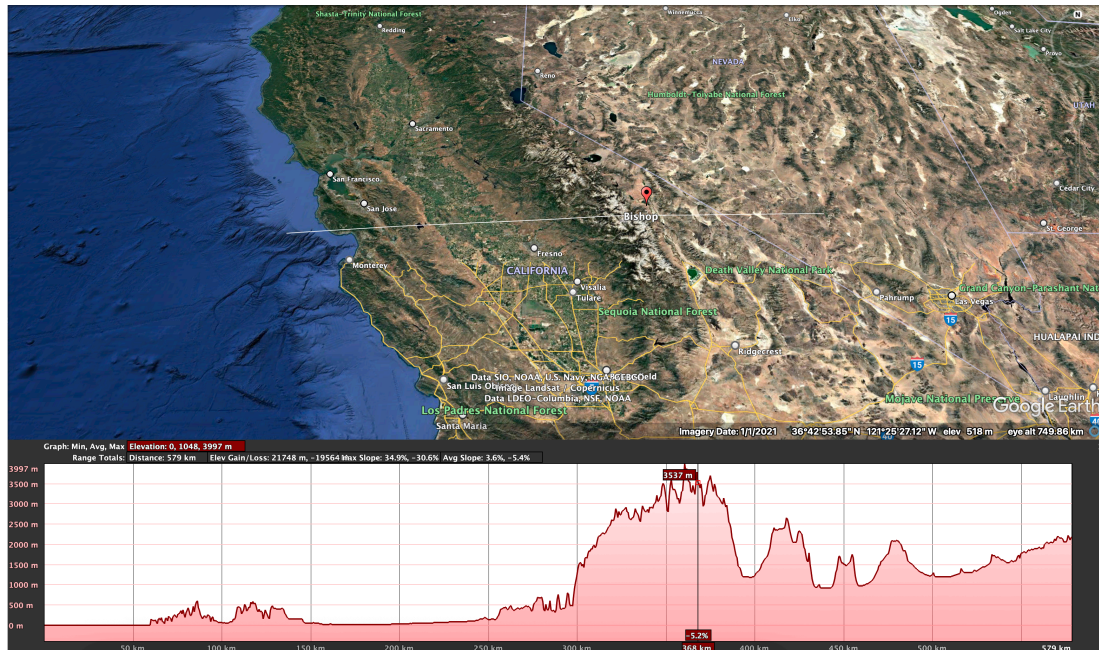


Figure 1: Sierra Nevada Transect on Google Earth

- Annotate approximate temperature changes ($^{\circ}\text{C}$) of a rising unsaturated parcel that moves:
 1. sea-level to 3 000 m.
 2. crest to Owens Valley floor. (Use the dry- and saturated-adiabatic lapse rates as appropriate.).
 3. (Use your online resources for this:) name **two rain-shadow communities** east of the crest and discuss, in ≈ 100 words, how this precipitation pattern shapes their water-supply strategies.

Paired Forest-Pasture Experiment (*fully hypothetical*)

Two tipping-bucket rain gauges operated for 14 consecutive days (Day 1 – Day 14):

- **Forest-Gauge** mounted 1 m above ground beneath a closed-canopy, mature forest.
- **Pasture-Gauge** sited in an open cattle pasture 600 m away on level terrain.

Day	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Σ 14-d
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Pasture-Gauge (mm)	0	5	12	0	20	8	4	15	9	22	7	17	11	10	140 mm
Forest-Gauge (mm)	0	4	9	0	17	7	3	13	8	18	6	15	9	10	119 mm

Table 1: Daily totals from the 14-day paired-gauge experiment.

1. Draw a **paired bar chart** comparing daily rainfall at the two gauges. The spreadsheet containing the data can be found [here](#).
2. The Forest-Gauge shows a **15 % lower** 14-day cumulative total than the Pasture-Gauge. Explain **three plausible micro-site or measurement biases** that could produce this difference.
3. For *each* bias, recommend one **practical adjustment or field check** that would reduce its impact in future campaigns.

Method Evaluation

A $2,500 \text{ km}^2$ catchment in central **Nepal** has four long-term rain gauges, all below $1,800 \text{ m asl}$. Snowfields begin at $\approx 3,000 \text{ m}$ and the crest reaches $5,000 \text{ m}$.

- Select **one** of the following to estimate areal rainfall for monthly water-balance work:
 - Thiessen polygons
 - Hypsometric (altitude-belt) method
- Justify your choice with reference to topography, gauge distribution, and expected precipitation gradients.
- Suggest **one supplementary data source/technology** that could improve the estimate without installing new high-altitude gauges, and explain how you would integrate it.